Whole-Home Wi-Fi Expansion: Option Analysis

RT-BE92U network: AiMesh node vs. third-party mesh, UniFi and GL.iNet Flint

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RT-BE92U AiMesh node to add	**$249.99** MSRP for one unit	**10G or 2.5G** Ethernet backhaul; wireless as fallback
2,882 Mb/s Host PHY ceiling at 160 MHz	**5 of 15** Options that pass every must (§7)	**V1–V8** Field checks; worksheet in Appendix A

Facts about this network are as stated, not re-measured; vendor figures are as listed; prices are listing snapshots (§13).

Contents

1 Verdict	1	8 Target architecture	9
2 Baseline network	2	9 Deployment plan	10
3 Requirements	2	10 Verification plan	10
4 AiMesh node candidates	3	11 Risks and unknowns	11
5 Alternatives outside AiMesh	3	12 What would change the verdict	12
6 Backhaul engineering	5	13 Method and abbreviations	12
7 Decision matrix	7	A Field worksheet	13

1 Verdict

Buy one ASUS RT-BE92U and add it as an AiMesh node on Ethernet backhaul: 10G when the primary's 10G port is free and the run is Cat6A (Cat6 is assured only to 37 m), otherwise 2.5G. Use MoCA over coax when no Ethernet run exists, and wireless MLO backhaul only when neither exists.

Confidence. High that an AiMesh node is the right fix: it follows from the stated configuration and vendor specifications. Medium on throughput: no RT-BE92U pair has been measured, so check V2 (§10) settles it.

1.1 Why this option

- Only AiMesh nodes extend coverage inside the primary's own roaming domain while leaving its services untouched: DoT to AdGuard, plaintext DHCP DNS, the USB SSD Samba share and the nvram configuration.
- The node carries the 6 GHz-only SSID the GTR9 Pro joins.
- The primary's 2×2 5 GHz and 6 GHz radios cap wireless backhaul at 2 spatial streams per band, so a costlier node adds at most receive diversity at range.
- It is cheaper than the other USD-priced tri-band node: $249.99 against $279.99 for the ZenWiFi BT8, which Amazon sold at $189.99 on 2026-04-27. No controller or PoE is needed.

1.2 Do not buy

- **ZenWiFi BD4/BD5:** dual-band, with no 6 GHz radio.
- **ZenWiFi BN12:** Wi-Fi 8 at £529.99, a UK pre-order with no ETA; its gains need Wi-Fi 8 clients, and US sale needs FCC conditional approval (U12).
- **Deco, Orbi or eero kits:** they cannot keep the Asuswrt services and one roaming domain without discarding the RT-BE92U's radios (§5.1).
- **Ubiquiti UniFi:** the same 2×2 class on 5 and 6 GHz, plus a controller and PoE for ceiling APs; justified only by a VLAN-driven rebuild (§5.2).
- **GL.iNet Flint 3 and Flint 4:** routers, not nodes; as a second access point they split roaming (§5.3), and the Flint 4 needs FCC conditional approval for US sale (U12).

2 Baseline network

Element	Current state	Consequence for this decision
Router	ASUS RT-BE92U on stock Asuswrt, by choice; Asuswrt-Merlin supports the unit but is not used	The node must be an ASUS AiMesh unit on stock firmware
Router ports	10G WAN/LAN and 2.5G WAN/LAN, both auto-sensing, plus 3 × 2.5G LAN	Wired backhaul at 10G or 2.5G (§6.1)
Router radios	2.4 GHz 3×3; 5 GHz 2×2; 6 GHz 2×2; channels up to 320 MHz	Wireless backhaul capped at 2 spatial streams per band
Link aggregation	Not offered on the firmware reviewed	Above 2.5G needs the 10G port at both ends
Upstream DNS	AdGuard ad-block tier as WAN DNS; router DoT on since 2026-07-31; AdGuard validates DNSSEC	Stays on the primary
Client DNS	Plaintext DNS from the router via DHCP, by design; the host pins no upstreams, DoT or DNSSEC settings	The node must not touch DHCP or DNS; no host-side DoT
Priority client	GTR9 Pro: MediaTek MT7925 (Filogic 360, 2×2 EHT) with a 160 MHz hardware ceiling; joins the 6 GHz-only SSID ASUS_68-6G at 160 MHz EHT	The node needs a 6 GHz radio; 320 MHz and Wi-Fi 8 headroom is unusable here
Storage	Lexar ES3 ext4 USB SSD on the router, shared over Samba	Stays on the primary
Segmentation	Smart Home Master sets up to three SSIDs; ASUS’s VLAN FAQ omits the RT-BE92U; Merlin adds no VLAN on this class	VLAN needs are the one case for a platform change (§12)
Wi-Fi security	WPA/WPA2-Enterprise is unavailable in AiMesh mode	Blocks the plan only if Enterprise authentication is in use (U3)

3 Requirements

Must-level requirements gate the options; should-level requirements rank those that pass.

ID	Requirement	Level	Rationale
R1	Extend 6 GHz coverage while carrying the host’s 6 GHz-only SSID	Must	The priority client uses 6 GHz only
R2	Keep the primary’s Asuswrt stack: DoT and AdGuard, DHCP DNS, USB SSD share, nvram configuration	Must	Deliberate design
R3	One roaming domain: a single system steering clients between access points	Must	AiMesh uses ASUS’s own roaming and band steering, described as proprietary rather than 802.11k/v/r-based; separate systems steer independently
R4	Backhaul of 2.5 Gb/s or more: Ethernet link rate or measured wireless throughput	Should	Keeps backhaul at or above realistic host goodput; host PHY ceiling 2,882 Mb/s (§13)
R5	No controller, PoE injectors or extra appliances	Should	Operational simplicity
R6	Low incremental cost	Should	A single-unit purchase is preferred

4 AiMesh node candidates

Node	6 GHz radio	Wired backhaul	Notes	Price snapshot (US unless noted)	Verdict
RT-BE92U	Yes, 2×2	10G or 2.5G	Identical to the primary: same radios, ports and firmware line	MSRP $249.99	**Recommended**
ZenWiFi BT8	Yes, tri-band	2.5G: one WAN, one LAN	Upright, 18.6 cm tall; seven internal antennas; MLO backhaul	MSRP $279.99; Amazon low $189.99 (2026-04-27), average $271.54	Alternate
ZenWiFi BT10	Yes, 4×4; 11,529 Mb/s	10G: 10G WAN/LAN and 10G LAN	Second 10G port; extra streams unusable against the primary's 2×2 radios	SG: S$1,076 per 2-pack	Overspend
ZenWiFi BD4/BD5	No, dual-band	BD5: 2.5G	Cannot carry the 6 GHz-only SSID; BD5 wireless backhaul ≈265 Mb/s	Not checked	Reject
ZenWiFi BN12	Yes, Wi-Fi 8; 11,529 Mb/s	10G: 2 × 10G, 3 × 2.5G	Joins AiMesh with most existing ASUS routers; Wi-Fi 8 gains need Wi-Fi 8 clients	UK: £529.99, pre-order, no ETA	Defer

Matched pair. A second RT-BE92U shares the primary's radios, ports and firmware line; the cost is that a model-specific firmware regression reaches both units at once (U5).

5 Alternatives outside AiMesh

5.1 Third-party mesh kits

TechRadar still rates the TP-Link Deco BE63 its top mesh system (guide updated 2026-02-17), a sound pick for a from-scratch install. On this network, any third-party kit hits one of two dead ends:

Integration	Consequence	Fails
Kit replaces the RT-BE92U	The DoT and AdGuard chain, the nvram configuration and the USB SSD share are rebuilt on the kit's platform or dropped	R2
Kit in AP mode behind the RT-BE92U	Two Wi-Fi systems steering independently, or the RT-BE92U's Wi-Fi 7 radios switched off and replaced with purchased ones	R3 or R6

Verdict: reject. Kits were not priced, because the only variant that meets R2 and R3 discards the RT-BE92U's own Wi-Fi 7 radios (R6).

5.2 Ubiquiti UniFi

Ubiquiti's consumer AmpliFi Alien is listed as discontinued since 2023-06-15; the Wi-Fi 7 gateways found for this report, the Express 7 and the Dream Router 7, are UniFi devices.

Device	Radios	Ports and power	Role	Price snapshot
UniFi Express 7	2×2 on 2.4, 5 and 6 GHz; 5,700 Mb/s at 6 GHz	10G WAN, one 2.5G LAN; USB-C 5 V/5 A power	Gateway with a controller for 30 devices, wired AP or wireless-backhaul mesh AP	$219.99 and $244.99 at one US retailer
UniFi U7 Pro	2×2 per band; 5,765 Mb/s at 6 GHz	One 2.5G port; PoE+ only, no power input; up to 21 W	Ceiling AP; standalone mode limits features	MSRP $200
UniFi Dream Router 7	2×2 on 2.4, 5 and 6 GHz	10G SFP+ and 2.5G WAN; 3 × 2.5G LAN including one PoE port; 15.4 W budget	Gateway with a controller; VLAN segmentation; microSD for camera and call recordings	Not checked (US)

Mode	What it takes	Outcome	Fails
Standalone APs, no controller	U7 Pro plus a PoE+ injector per AP	No 802.11r/k/v roaming between APs	R3, R5
Managed APs behind the RT-BE92U	A UniFi console or the self-hosted UniFi Network Server (Windows, macOS, Linux); PoE+ per AP	Two Wi-Fi systems unless the RT-BE92U's radios are switched off	R3 or R6; R5
Full swap: UX7 or UDR7 plus APs	Rebuild DNS: UniFi Encrypted DNS takes a DNS-over-HTTPS stamp, the way AdGuard's own UniFi guide sets it; DoT is not documented. No USB storage share: RTINGS reports no USB port on the UDR7 (conflict: U8). The UDR7's 15.4 W PoE cannot feed a U7 Pro drawing up to 21 W	One roaming domain plus VLAN segmentation, which ASUS does not list for the RT-BE92U	R2

Verdict: lateral on radios (2×2 on 5 and 6 GHz, like the RT-BE92U), heavier to run and destructive to the Asuswrt stack in its only roaming-clean form. Revisit only if VLANs become a requirement that Smart Home Master SSIDs cannot meet (§12).

5.3 GL.iNet Flint 3 and Flint 4

GL.iNet's Flint routers run OpenWrt-based firmware on a Qualcomm (Flint 3) or MediaTek (Flint 4) platform. They are routers, not AiMesh nodes: on this network a Flint can only run as a second Wi-Fi system behind the RT-BE92U or replace it.

Device	Radios	Ports and USB	Firmware and modes	Price snapshot
Flint 3 (GL-BE9300)	2×2 on all bands; 688 / 2,882 / 5,765 Mb/s; MLO; RTINGS: good range, speed suits a 2 Gb/s connection	Five 2.5G ports: WAN, WAN/LAN and three LAN; USB 3.0; no 10G	OpenWrt 23.05 fork, admin panel 4.x, AdGuard Home; Router, Access Point, Extender and WDS modes, with the wired web GUI unreachable in AP or WDS mode	$189.90 direct; $143 to $158 via GL.iNet's AliExpress store
Flint 4 (GL-BE14000)	688 / 4,323 / 8,646 Mb/s; six antennas; MLO behavior unconfirmed	10G RJ45 and 10G SFP+ WAN/LAN, 2.5G WAN/LAN, 3 × 2.5G LAN, 4 × 1G LAN; USB 3.0 Type-A and Type-C	OpenWrt-based; MediaTek quad-core A73, 2 GB RAM, 64 GB eMMC; VPN up to 1.5 Gb/s	$299 pre-order (2026-08-04), GL.iNet store only; needs US approval (U12)

Fit against the Asuswrt stack (R2)

Asuswrt service	GL.iNet equivalent	Fit
Router DoT to AdGuard DNS	Encrypted DNS mode offers DNS over TLS and DNS over HTTPS; a custom server such as a private AdGuard DNS endpoint needs SSH and is not officially supported	Partial
USB SSD Samba share	The File Sharing app serves USB storage over Samba (SMB2); both Flints have USB 3.0	Yes
Ad blocking	AdGuard Home runs on the router	Yes
Segmentation	An isolated guest network since firmware 4.5; full VLANs on the Flint 3 only through LuCI or UCI scripts, which community authors warn can crash-loop the router	VLANs, no GUI
AiMesh roaming domain	"Mesh" appears in the Flint 3 specifications; the roaming method and partner models are undocumented (U13)	Unknown
nvram configuration	OpenWrt UCI; nothing carries over	Rebuild

Mode	What it takes	Outcome	Fails
Flint 3 as AP or extender behind the RT-BE92U	Cable to a 2.5G LAN port; the Flint 3 has no 10G port	A second Wi-Fi system steering on its own; wired GUI access lost in AP or WDS mode	R3
Flint 3 or 4 replaces the primary (RT-BE92U becomes an AP)	Rebuild DoT (custom server via SSH), Samba, DHCP and firewall; VLANs via UCI	Coverage still comes from a second, independent system, and the Asuswrt stack is gone	R2, R3
Flint pair: router plus extender or mesh	2 × Flint 3 at $189.90	One vendor domain; roaming method unverified (U13)	R2; R6
Flint 4 in any role	A model announced after 2026-03-23	US sale needs FCC conditional approval (U12)	Availability

Verdict: reject for Wi-Fi expansion. A Flint adds coverage only as a second, independently steering system (R3) or by replacing the primary (R2). Keep the Flint 3 in view as the cheaper VLAN-capable rebuild: $189.90, with DoT, Samba and ad blocking built in and VLANs by command line rather than GUI (§12).

6 Backhaul engineering

6.1 Paths

Choose the path from what the house already has (Figure 1), then free the port the primary needs for it (Figure 2). Figure 3 sets the resulting ceilings against the only wireless backhaul figures that were measured.

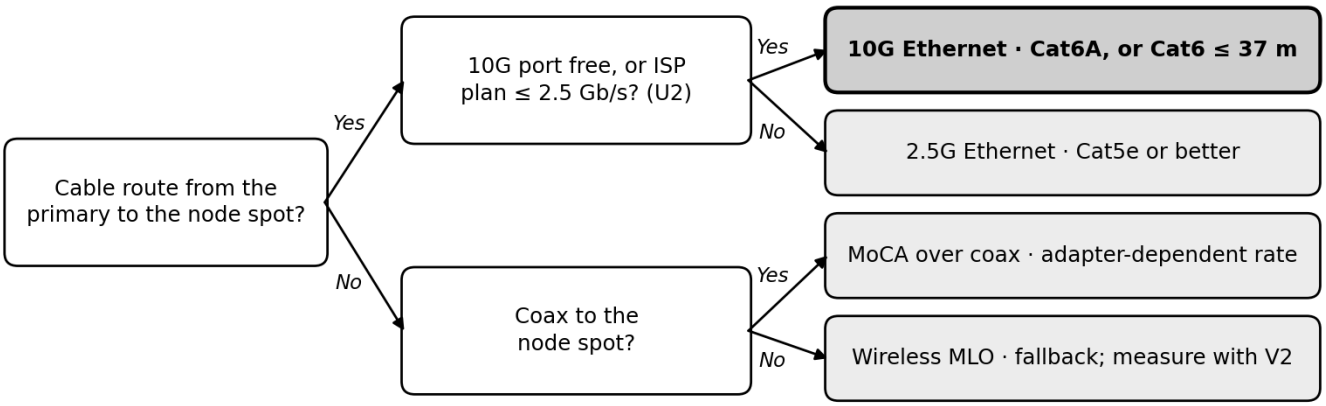


Figure 1. Backhaul path selection. The 10G branch needs the primary's 10G port free (Figure 2); Cat6 is assured only to 37 m and reaches 55 m only where alien crosstalk allows.

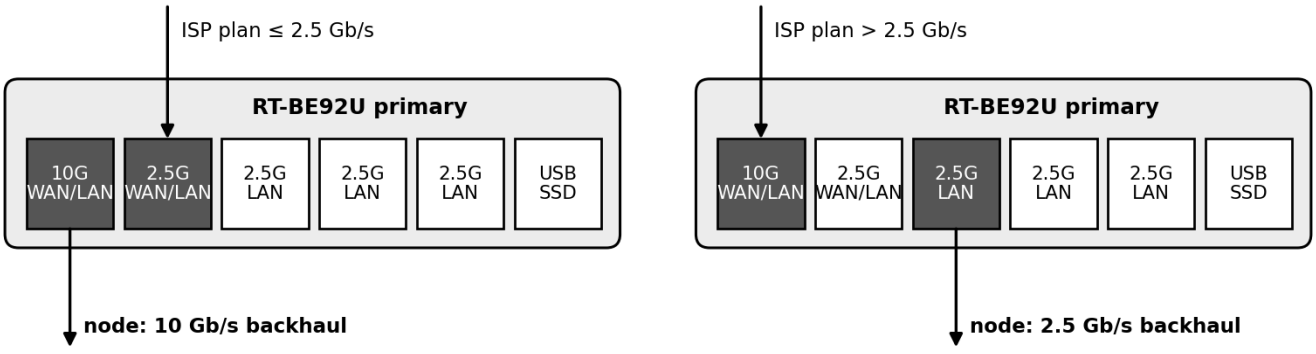


Figure 2. Port plan on the primary (schematic; physical port order not shown). Left: an ISP plan of 2.5 Gb/s or less moves WAN to the 2.5G WAN/LAN port and frees 10G for the node. Right: a faster plan keeps WAN on 10G, and a 2.5G LAN port feeds the node.

Path	Ceiling	Cabling	Notes
10G Ethernet	10 Gb/s link rate	Cat6A to 100 m; Cat6 55 m typical, 37 m under worst-case bundling	Needs the primary's 10G port free (Figure 2)
2.5G Ethernet	2.5 Gb/s link rate	Cat5e or Cat6 to 100 m	Any LAN port; under the host's 2,882 Mb/s PHY ceiling, but real Wi-Fi goodput runs well below PHY
MoCA over coax	Adapter-dependent (not assessed)	Existing coax	Wired-like option; one owner runs AiMesh nodes on MoCA behind an RT-BE92U
Wireless MLO, 5 and 6 GHz	Measured: BT10 pair ≈727 Mb/s average, ≈1 Gb/s peak; BD5 pair ≈265 Mb/s	None	The node's single 6 GHz radio carries the host link and the backhaul leg (§6.2)

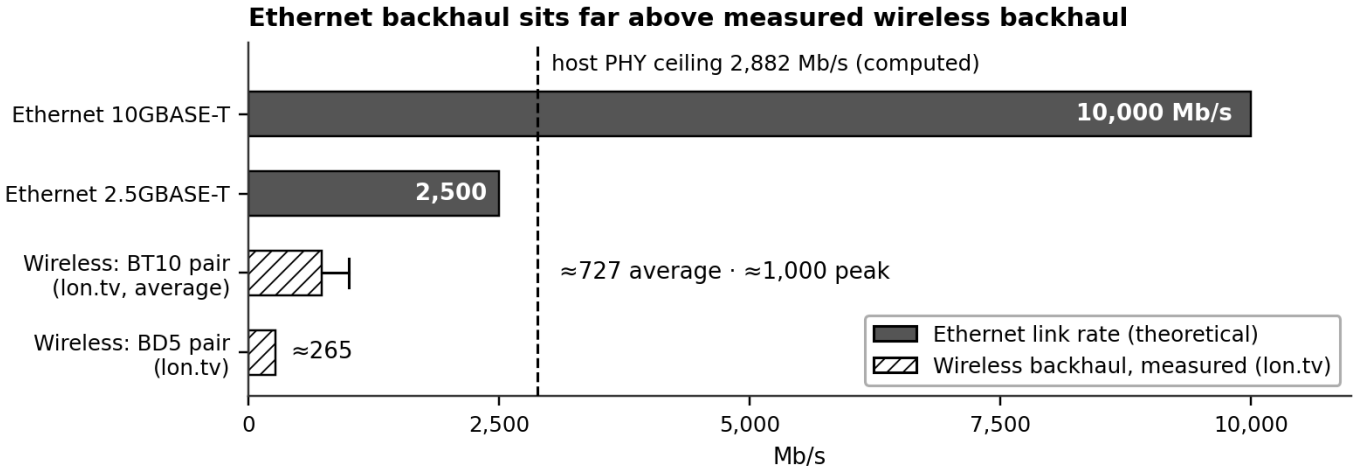


Figure 3. Backhaul ceilings. Solid: Ethernet link rates (theoretical). Hatched: lon.tv's measured ZenWiFi wireless backhaul; no RT-BE92U pair was measured. Dashed: host PHY ceiling (computed).

6.2 Airtime on tri-band units

Each tri-band unit here has one 6 GHz radio. With wireless backhaul, the node's 6 GHz radio carries both the host's link and the backhaul leg on the same channel, so host traffic crosses that channel twice unless MLO moves the backhaul to 5 GHz. Wired backhaul removes the second crossing (Figure 4).

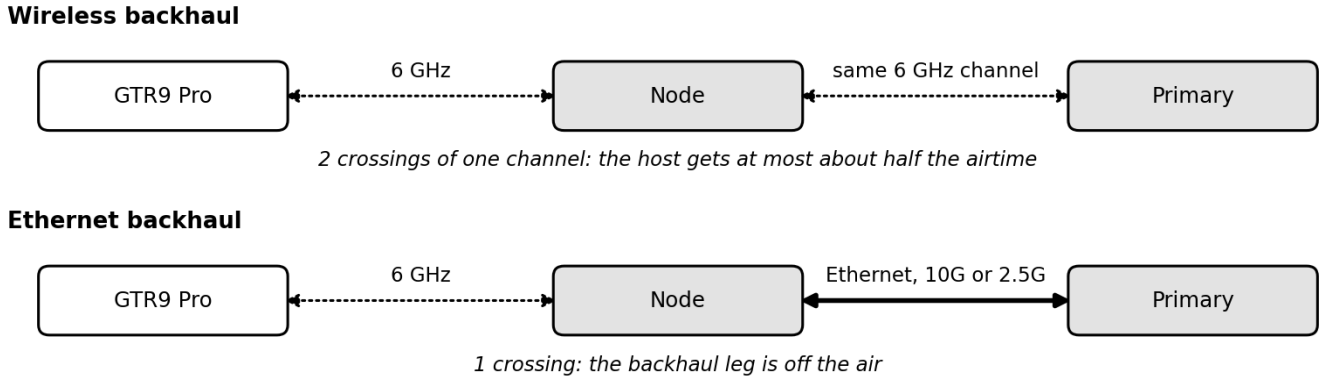


Figure 4. Airtime on a one-radio node (conceptual). With wireless backhaul each host frame crosses the 6 GHz channel twice, so the host gets at most about half the channel's airtime (inference); Ethernet takes the second hop off the air.

6.3 Headroom against the host ceiling

Figure 5 rates every candidate radio against the host. Capacity right of the dashed line is unusable by the GTR9 Pro, whose MT7925 tops out at 160 MHz.

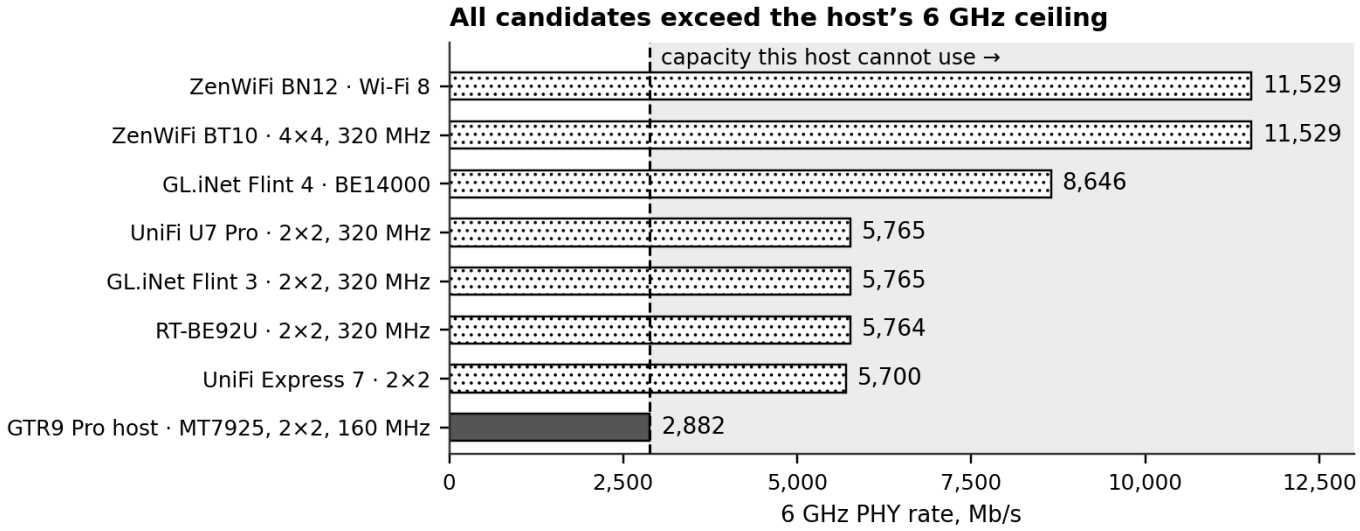


Figure 5. 6 GHz PHY ceilings, vendor-rated as listed by ASUS, GL.iNet and retailers; host value computed for EHT, 2 spatial streams, 160 MHz, MCS 13, 0.8 µs GI (§13).

7 Decision matrix

Scored against §3: PASS, PART and FAIL are judgments from the facts above; n/c means not checked. Figure 6 ranks the options; the matrix that follows shows each judgment, with shading that repeats the label for black-and-white print.

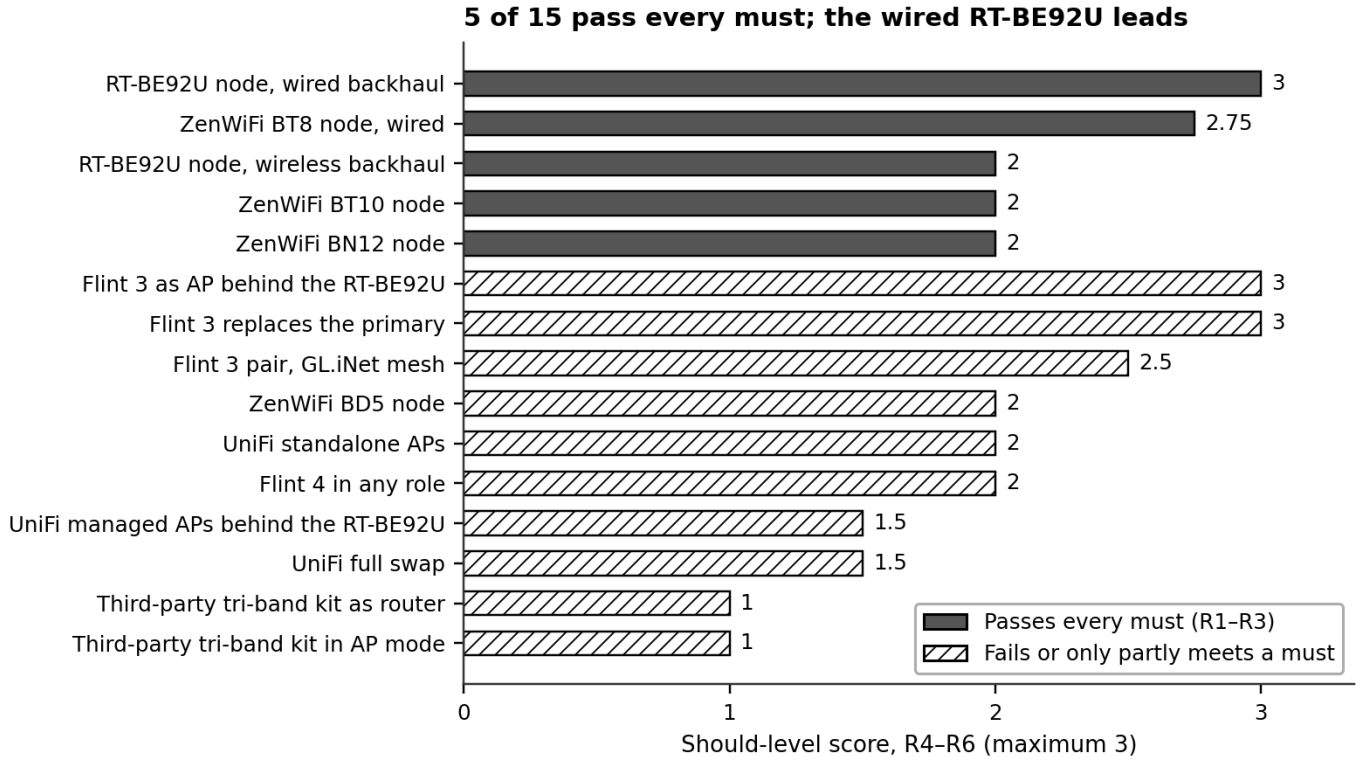


Figure 6. Should-level score (R4–R6) per option: PASS 1, PART ½, FAIL or n/c 0; cost LOW 1, LOW+ ¾, MED ½, HIGH or n/c 0. Hatched options fail or only partly meet a must-level requirement (R1–R3) and are out regardless of score.

Option	R1 6 GHz	R2 stack	R3 roam	R4 ≥2.5G	R5 no extra gear	R6 cost	Verdict
RT-BE92U node, wired backhaul	**PASS**	**PASS**	**PASS**	**PASS**	**PASS**	LOW	**Recommended**
RT-BE92U node, wireless backhaul	**PASS**	**PASS**	**PASS**	**FAIL**	**PASS**	LOW	Fallback
ZenWiFi BT8 node, wired	**PASS**	**PASS**	**PASS**	**PASS**	**PASS**	LOW+	Alternate
ZenWiFi BT10 node	**PASS**	**PASS**	**PASS**	**PASS**	**PASS**	HIGH	Overspend
ZenWiFi BD5 node	**FAIL**	**PASS**	**PASS**	**PASS**	**PASS**	*n/c*	Reject
ZenWiFi BN12 node	**PASS**	**PASS**	**PASS**	**PASS**	**PASS**	HIGH	Defer
Third-party tri-band kit as router	**PASS**	**FAIL**	**PASS**	*n/c*	**PASS**	*n/c*	Reject
Third-party tri-band kit in AP mode	**PASS**	**PASS**	**PART**	*n/c*	**PASS**	*n/c*	Reject
UniFi standalone APs	**PASS**	**PASS**	**FAIL**	**PASS**	**FAIL**	LOW	Reject
UniFi managed APs behind the RT-BE92U	**PASS**	**PASS**	**PART**	**PASS**	**FAIL**	MED	Reject
UniFi full swap	**PASS**	**FAIL**	**PASS**	**PASS**	**PART**	HIGH	VLAN rebuild only
Flint 3 as AP behind the RT-BE92U	**PASS**	**PASS**	**PART**	**PASS**	**PASS**	LOW	Reject
Flint 3 replaces the primary	**PASS**	**FAIL**	**PART**	**PASS**	**PASS**	LOW	Cheaper VLAN rebuild
Flint 3 pair, GL.iNet mesh	**PASS**	**FAIL**	*n/c*	**PASS**	**PASS**	MED	Reject
Flint 4 in any role	**PASS**	*n/c*	*n/c*	**PASS**	**PASS**	HIGH	Needs US approval

Cost tokens are relative: LOW ≈ one RT-BE92U (MSRP $249.99); LOW+ slightly more; MED adds controller or PoE gear; HIGH multiplies unit cost. R4 for wireless backhaul is inferred from lon.tv's ZenWiFi measurements; the BN12 row rests on its announced AiMesh support; Flint rows follow the roles GL.iNet documents (§5.3). Figure 7 lists the price snapshots behind the cost tokens and Figure 8 sets price against the 6 GHz ceiling.

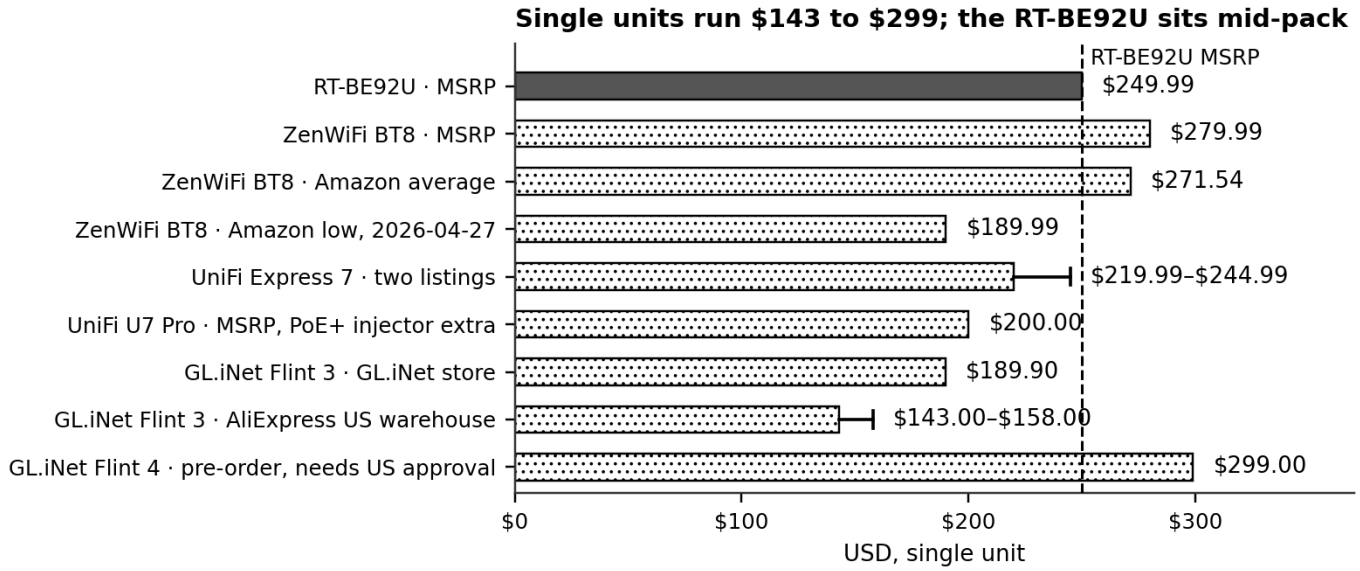


Figure 7. USD price snapshots for single units: SHI MSRPs, camelcamelcamel's Amazon history, Altex listings, GL.iNet's store and AliExpress listings, and the Flint 4 pre-order. Listings in S$ and £ (BT10, BN12) are omitted rather than converted.

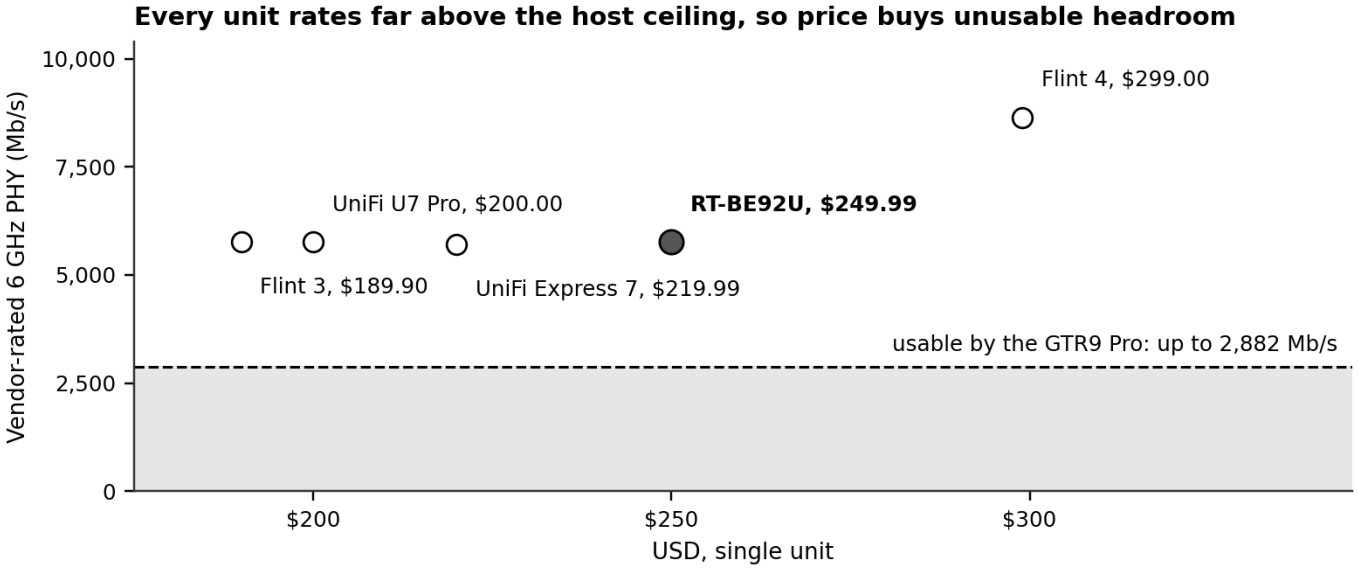


Figure 8. Single-unit price against the vendor-rated 6 GHz PHY, for the options with a US dollar price. Roles differ: the U7 Pro is a ceiling AP needing PoE+, the Flints are routers (§5). The ZenWiFi BT8 is omitted because ASUS lists no separate 6 GHz rate for it; the BT10 and BN12 are omitted for want of a US price.

8 Target architecture

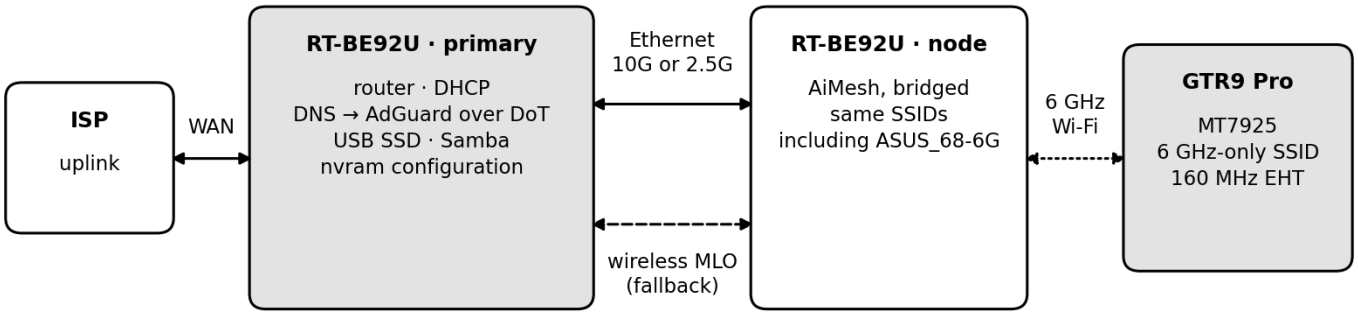


Figure 9. Target topology. Dashed: wireless MLO backhaul, fallback only.

The primary keeps routing, DHCP, DNS (AdGuard over DoT) and the Samba share; the node only bridges, and the host keeps plaintext DHCP DNS as designed. AiMesh roams clients across nodes; checks V1 and V4 verify that the node carries ASUS_68-6G (Figure 9).

9 Deployment plan

Step	Action	Done when
P0 Baseline	Record V1–V3 at the weak spot, V5–V7 anywhere and the primary's firmware version (Appendix A)	Values saved; V1's BSSID test waits for P5
P1 Backhaul path	Confirm an Ethernet or coax route (U1) and the WAN port (U2); pick a panel of Figure 2	Path and ports chosen
P2 Procure	One RT-BE92U (MSRP $249.99); Cat6A for 10G, Cat5e or better for 2.5G	Unit and cable on hand
P3 Prepare the node	Update it to the primary's stock release line, then factory-reset it for pairing	Node at defaults, same release line
P4 Protect the primary	Save the configuration before any change (nvram save); it lives only in nvram and is lost on factory reset	Backup stored off the router
P5 Pair	Add the node from the AiMesh page or the ASUS Router app; cable the chosen primary port to the node's 10G or 2.5G WAN/LAN port	Node listed in AiMesh
P6 Pin the backhaul	Set the node's Backhaul Connection Priority (AiMesh → Topology → node → Management) to "Ethernet first", which falls back to wireless; once V8 shows Ethernet and airtime matters more than fallback, switch to Ethernet Backhaul Mode (AiMesh → System Settings)	V8 shows Ethernet backhaul
P7 Verify	Re-run V1–V8 at the P0 spots	Every pass criterion met
P8 Roll back	Remove the node from AiMesh; restore the P4 backup if a primary setting changed	Primary matches P0

10 Verification plan

The first table maps each requirement of §3 to the checks that settle it. Commands run on the GTR9 Pro unless noted; values in angle brackets are placeholders, and `iw dev` lists the wireless interface name. None of these commands change host configuration. Record results in Appendix A.

Requirement (§3)	Field check	When	If it fails
R1 6 GHz coverage on the host's SSID	V1	P7	Re-site the node; confirm the SSID reaches it
R2 Asuswrt stack untouched	V5, V6, V7	P0 and P7	Roll back (P8); the node must not serve DHCP or DNS
R3 One roaming domain	V4	P7	Check both units are in AiMesh, not bridged
R4 Backhaul of 2.5 Gb/s or more	V2, V8	P7	Re-seat the cable; re-check the port pair (Figure 2)
R5 No controller or extra gear	none; set at purchase	P2	Not applicable to an AiMesh node
R6 Low incremental cost	none; set at purchase	P2	Re-check prices (U6)

ID	Check	Command	Pass criterion
V1	6 GHz association via the node	`iw dev <iface> link`	freq between 5,925 and 7,125 MHz; tx bitrate shows "160MHz" and "EHT-MCS" (iw output); after P5 the BSSID near the node differs from the primary's

V2	Throughput, both directions	`# wired LAN host`
		`iperf3 -s`
		`# GTR9 Pro`
		`iperf3 -c <server> -t 30 -P 4`
		`iperf3 -c <server> -t 30 -P 4 -R`
			Both directions at or above the P0 baseline at the same spot

V3	Latency and loss to the primary	`ping -c 100 <primary-lan-ip>`	0% loss; avg and mdev at or below the baseline
V4	Roaming	`iw event -t`	Walk from the primary to the node and back: a new association to the node's BSSID appears; any disconnect-to-association gap in the timestamps is under 1 s
V5	DNS invariant	`resolvectl status <iface>` `resolvectl query example.com`	DNS server is the primary's LAN IP from DHCP; the link shows -DNSOverTLS; the query resolves; identical to P0

ID	Check	Command	Pass criterion
V6	USB share invariant	`smbclient -L //<primary-lan-ip> -U <user>`	Share listed, identical to P0 (needs the smbclient package)
V7	DHCP lease invariant	`ip -br addr show dev <iface>`	Address still in the primary's LAN subnet
V8	Backhaul type (router)	Primary web GUI → AiMesh → Topology → node; no CLI equivalent verified	The node reports Ethernet backhaul on a wired build

11 Risks and unknowns

ID	Item	Impact	Resolution	When
U1	Ethernet or coax path to the node location	Decides wired or wireless backhaul	Survey the route; MoCA if only coax exists	Before buying
U2	Which RT-BE92U port carries WAN; ISP plan rate	Decides 10G or 2.5G backhaul	Port plan (Figure 2)	Before buying
U3	WPA/WPA2-Enterprise in use	AiMesh mode disables it	Confirm it is unused; otherwise an AP-based design	Before buying
U4	Wireless-fallback airtime	Host link and backhaul leg share the node's 6 GHz radio (§6.2)	Prefer wired; quantify with V2	At install
U5	Firmware	One release line keeps AiMesh in lockstep, but a model-specific regression reaches both units	Record versions at P0 and P7; defer non-security updates briefly; keep the P4 backup	Ongoing
U6	Prices	Listing snapshots	Re-check at purchase	Before buying
U7	BT8 port conflict	ASUS lists dual 2.5G; one reseller lists dual 10G and 18 Gb/s	The ASUS specification governs	Only if BT8
U8	UniFi storage share	No SMB share feature identified; RTINGS reports no USB port on the UDR7, while one retailer lists USB 3.0	Matters only for a full swap; confirm on ui.com before buying	Only if UniFi
U9	BN12 and Wi-Fi 8	No ETA at Scan; Pokde.Net reports an October launch, Dong Knows reports 2027; Wi-Fi 8 gains need Wi-Fi 8 clients, and the final IEEE specification can change features; US sale needs conditional approval (U12)	Revisit when Wi-Fi 8 clients exist and it is on US sale	Later
U10	Other clients	Phone and laptop Wi-Fi not assessed	Node coverage helps them; verdict unchanged	Later
U11	Segmentation without a platform change	Smart Home Master sets up to three SSIDs; isolation not assessed	Test before treating VLAN needs as a reason to switch (§12)	Later
U12	US router rule (FCC Covered List)	Since 2026-03-23, new foreign-made consumer router models cannot be authorized for US import or sale without conditional approval; models approved earlier stay on sale, with update waivers to at least 2027-03-01, reportedly extended to January 2029. GL.iNet says its US-sold models are unaffected; the BN12 and Flint 4 were announced after the cut-off	Buy models already on US sale (RT-BE92U, BT8, Flint 3); check the FCC ID of any model announced after March 2026	Before buying
U13	GL.iNet mesh and roaming	"Mesh" is listed for the Flint 3; roaming method and partner models undocumented	Verify before any Flint pair	Only if Flint
U14	Flint DoT to a private AdGuard DNS endpoint	Custom DoT or DoH servers need SSH and are not officially supported	Treat as rebuild cost; test on firmware 4.8.x	Only if Flint

12 What would change the verdict

- **A cable reaches the GTR9 Pro’s location:** wire the host directly; the node then serves the other clients.
- **Segmentation becomes a requirement:** test Smart Home Master SSIDs first (U11); if true VLANs are needed, the cheaper rebuild is a Flint 3 (VLANs through LuCI or UCI; DoT, Samba and ad blocking built in, §5.3) and the managed one is UniFi (§5.2); both discard the R2 stack.
- **Enterprise Wi-Fi authentication becomes necessary:** AiMesh is out; move to an AP-based design.
- **The node must sit in a living space:** ZenWiFi BT8 instead, accepting its 2.5G wired ceiling.
- **Wi-Fi 8 clients arrive and the BN12 is on US sale:** re-evaluate with a Wi-Fi 8 primary; Multi-AP Coordination ties router and nodes, so a Wi-Fi 7 primary would gate it (inference).

13 Method and abbreviations

Facts about this network are as stated and were not re-measured. Vendor figures are as listed by manufacturers and retailers; reviews, listings and regulatory notices were checked from 2026-10-01 to 2026-10-03; this edition does not reproduce the source list. The host PHY rate is computed as 1,960 data tones × 12 bits × 5/6 ÷ 13.6 µs × 2 spatial streams = 2,882 Mb/s (EHT, 160 MHz, MCS 13, 0.8 µs GI); other PHY rates are vendor ratings. Prices are listing snapshots, not checkout quotes. Inferences are marked as such.

Abbreviations

Term	Meaning	Term	Meaning
AP	access point	**BSSID**	MAC identifier of one access point radio
DHCP	protocol that hands clients an address and DNS server	**DNSSEC**	signature validation for DNS answers
DoH	DNS over HTTPS	**DoT**	DNS over TLS
EHT	Extremely High Throughput (802.11be, Wi-Fi 7)	**GI**	guard interval between symbols
FCC	US Federal Communications Commission	**ISP**	internet service provider
LuCI, UCI	OpenWrt web interface and command-line configuration	**MCS**	modulation and coding scheme
MLO	Multi-Link Operation: one Wi-Fi 7 link across bands	**MoCA**	Multimedia over Coax Alliance networking
MSRP	manufacturer’s suggested retail price	**nvram**	router’s non-volatile settings store
PHY	physical layer; PHY rate is the raw link rate	**PoE, PoE+**	Power over Ethernet (802.3af, 802.3at)
SFP+	pluggable 10G transceiver port	**SMB**	Windows file-sharing protocol (Samba)
SSH	secure shell command-line access	**SSID**	Wi-Fi network name
VLAN	virtual LAN: a separated segment over shared wiring	**WDS**	Wireless Distribution System bridging

This edition. v1.6 (2026-10-03): contents with page numbers, §1.1 and §1.2, Figure 8, the requirement-to-check table, the risk timing column, and worksheet notes and outcome line. Earlier editions are listed in the repository CHANGELOG.

A Field worksheet

Print this page. Record each check at P0, before buying, and again at P7, after pairing; pass criteria are in §10. V4 and V8 have no P0 value: both need the node in place.

Date		**Node location**	
Primary firmware		**Node firmware**	
ISP plan		**Backhaul**	☐ 10G ☐ 2.5G ☐ MoCA ☐ wireless

ID	Check	Record	P0 baseline	P7 after	Pass
V1	6 GHz link via the node	freq; tx bitrate; BSSID			☐
V2	Throughput	Mb/s down / up			☐
V3	Latency and loss	loss %; avg / mdev ms			☐
V4	Roaming	longest gap (s)	*n/a*		☐
V5	DNS invariant	DNS server; DoT flag			☐
V6	USB share	share listed			☐
V7	DHCP lease	address and subnet			☐
V8	Backhaul type	Ethernet or wireless	*n/a*		☐

Notes: channel or width changes, BSSID flaps, firmware versions, dead spots

Outcome ☐ keep the node ☐ re-site it and re-run V1–V3 ☐ roll back (P8)